

Industrial Internship Report

Music Player with Playlist using JDBC

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Executive Summary

This report presents the development of a Music Player with Playlist Management using Java and JDBC. The application allows users to play music, manage playlists, store song information in a database, and retrieve records efficiently.

1. Preface

The internship provided practical exposure to software development, database integration, problem solving, and project implementation. The project was completed over six weeks.

2. Introduction

The project focuses on designing a desktop-based music player application integrated with a database through JDBC. The system manages songs, playlists, and playback controls.

3. Problem Statement

Develop a Music Player Application that allows users to play songs, create playlists, manage music collections, and store data using JDBC-based database connectivity.

4. Existing and Proposed Solution

Traditional music players mainly provide playback functionality. The proposed solution extends this by providing playlist management, database storage, search capabilities, and efficient retrieval of song information.

5. Proposed Design / Model

The system consists of User Interface, Music Management Module, Playback Module, and Database Module. JDBC is used to connect Java applications with the database.

System Features

Music Playback (Play, Pause, Stop, Skip)

Music Library Management

Playlist Creation and Management

Add and Remove Songs from Playlists

Search and Filter Songs

Database Storage using JDBC

User-Friendly Interface

Error Handling and Logging

Database Design

The database stores song information such as title, artist, album, genre, and duration. Playlist tables maintain mappings between songs and playlists.

6. Performance Testing

The application was tested for playback operations, database connectivity, playlist creation, modification, and search functionality.

Test Cases

- Verify song playback
- Verify playlist creation
- Verify song addition/removal
- Verify database retrieval

Performance Outcome

The system successfully executed all required operations with reliable database access and responsive playback controls.

7. My Learnings

I gained practical experience in Java programming, JDBC connectivity, database design, SQL queries, debugging, exception handling, and software testing.

8. Future Work Scope

Future enhancements may include cloud synchronization, recommendation systems, online streaming, user authentication, mobile application support, and advanced audio controls.